

TASK 1:

```
Ubuntu 24.04 (Snapshot 1) [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
Jan 29 15:34
osuser01@ubuntu: /home/ubuntu

Info: Adding group 'oslab01' (GID 1002) ...
ubuntu@ubuntu:~$ sudo adduser --ingroup oslab01 osuser01
Info: Adding user 'osuser01' ...
Info: Selecting UID from range 1000 to 59999 ...

Info: Adding new user 'osuser01' (1002) with group 'oslab01' (1002) ...
warn: The home directory /home/osuser01 already exists. Not touching this directory.
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for osuser01
Enter the new value, or press ENTER for the default
Full Name []: osfan
Room Number []:
Work Phone []:
Home Phone []:
Other []:

Is the information correct? [Y/n] y
Info: Adding new user 'osuser01' to supplemental / extra groups 'users' ...
Info: Adding user 'osuser01' to group 'users' ...
ubuntu@ubuntu:~$ sudo adduser --ingroup oslab01 osuser02
Info: Adding user 'osuser02' ...
Info: Selecting UID from range 1000 to 59999 ...

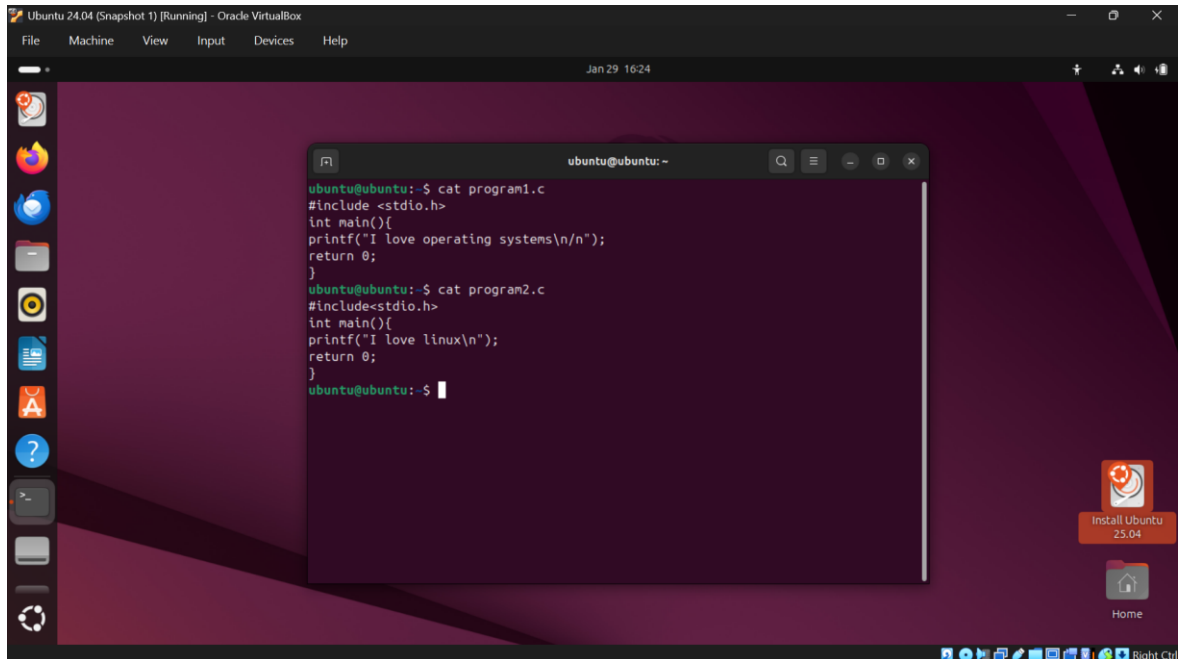
Info: Adding new user 'osuser02' (1003) with group 'oslab01' (1002) ...
warn: The home directory /home/osuser02 already exists. Not touching this directory.
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for osuser02
Enter the new value, or press ENTER for the default
Full Name []: shmed
Room Number []:
Work Phone []:
Home Phone []:
Other []:

Is the information correct? [Y/n] y
Info: Adding new user 'osuser02' to supplemental / extra groups 'users' ...
Info: Adding user 'osuser02' to group 'users' ...
ubuntu@ubuntu:~$ sudo addgroup sudo
fatal: The group 'sudo' already exists.
ubuntu@ubuntu:~$ sudo usermod -aG sudo osuser01
ubuntu@ubuntu:~$ sudo usermod -aG sudo osuser02
ubuntu@ubuntu:~$ sudo groups osuser01
osuser01 : oslab01 sudo users
ubuntu@ubuntu:~$ sudo groups osuser02
osuser02 : oslab01 sudo users
ubuntu@ubuntu:~$ su osuser01
Password:
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.
osuser01@ubuntu:~/home/oslab01$
```

TASK 2:

```
Ubuntu 24.04 (Snapshot 1) [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
Jan 29 16:22
ubuntu@ubuntu:~$ nano program1.c
ubuntu@ubuntu:~$ nano program2.c
ubuntu@ubuntu:~$ gcc -o Program1 program1.c
ubuntu@ubuntu:~$ gcc -o Program2 program2.c
ubuntu@ubuntu:~$ ./Program1
I love operating systems
/nubuntu@ubuntu:~$ ./Program2
I love linux
ubuntu@ubuntu:~$ ./Program1 > output1.txt
ubuntu@ubuntu:~$ ./Program2 > output2.txt
ubuntu@ubuntu:~$ cat output1.txt
I love operating systems
/nubuntu@ubuntu:~$ cat output2.txt
I love linux
ubuntu@ubuntu:~$ cat output1.txt output2.txt > OSLAB.txt
ubuntu@ubuntu:~$ cat OSLAB.txt
I love operating systems
/n/i love linux
ubuntu@ubuntu:~$
```

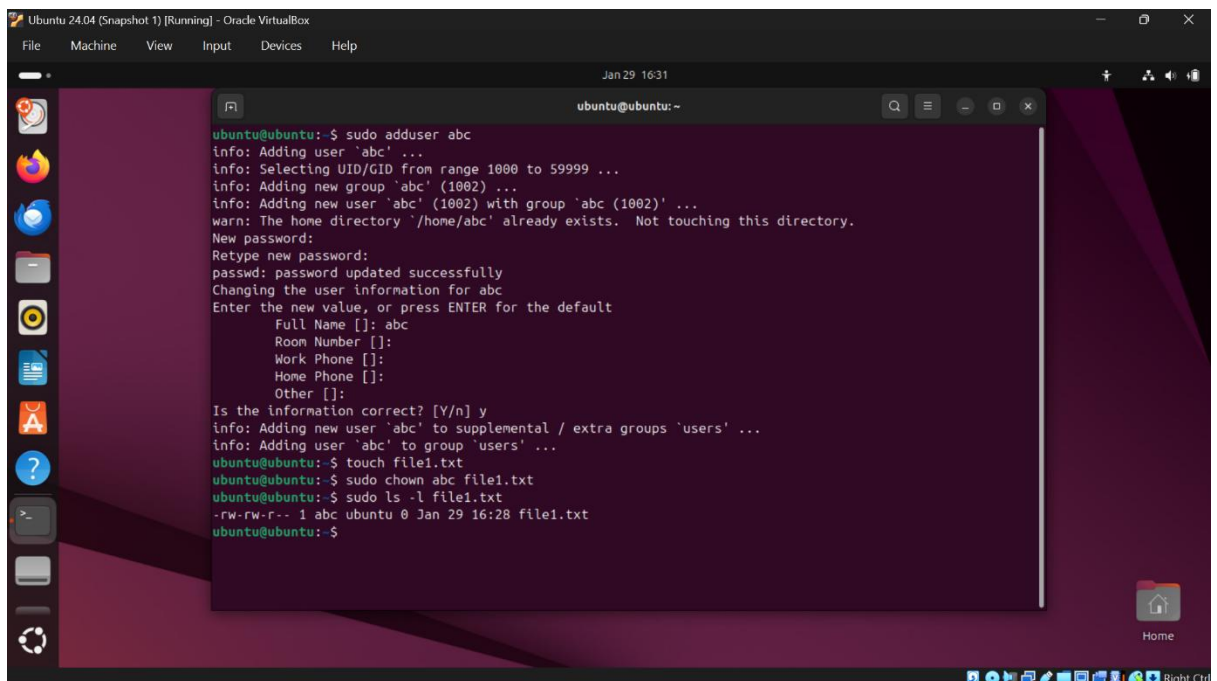
TASK 3:



The screenshot shows a terminal window titled 'ubuntu@ubuntu: ~' with the following commands and output:

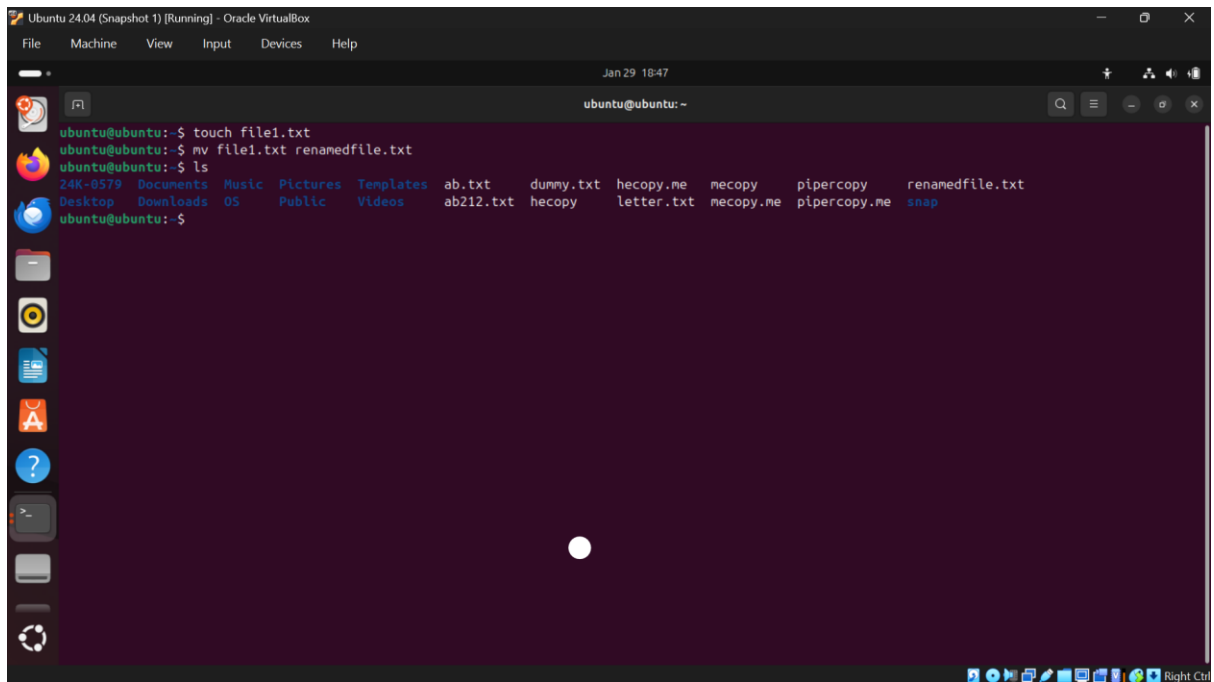
```
ubuntu@ubuntu:~$ cat program1.c
#include <stdio.h>
int main(){
    printf("I love operating systems\n\n");
    return 0;
}
ubuntu@ubuntu:~$ cat program2.c
#include<stdio.h>
int main(){
    printf("I love linux\n");
    return 0;
}
ubuntu@ubuntu:~$
```

TASK 4:



The screenshot shows a terminal window titled 'ubuntu@ubuntu: ~' with the following commands and output:

```
ubuntu@ubuntu:~$ sudo adduser abc
info: Adding user 'abc' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group 'abc' (1002) ...
info: Adding new user 'abc' (1002) with group 'abc (1002)' ...
warn: The home directory '/home/abc' already exists. Not touching this directory.
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for abc
Enter the new value, or press ENTER for the default
  Full Name []: abc
   Room Number []:
    Work Phone []:
    Home Phone []:
       Other []:
Is the information correct? [Y/n] y
info: Adding new user 'abc' to supplemental / extra groups 'users' ...
info: Adding user 'abc' to group 'users' ...
ubuntu@ubuntu:~$ touch file1.txt
ubuntu@ubuntu:~$ sudo chown abc file1.txt
ubuntu@ubuntu:~$ sudo ls -l file1.txt
-rw-rw-r-- 1 abc ubuntu 0 Jan 29 16:28 file1.txt
ubuntu@ubuntu:~$
```



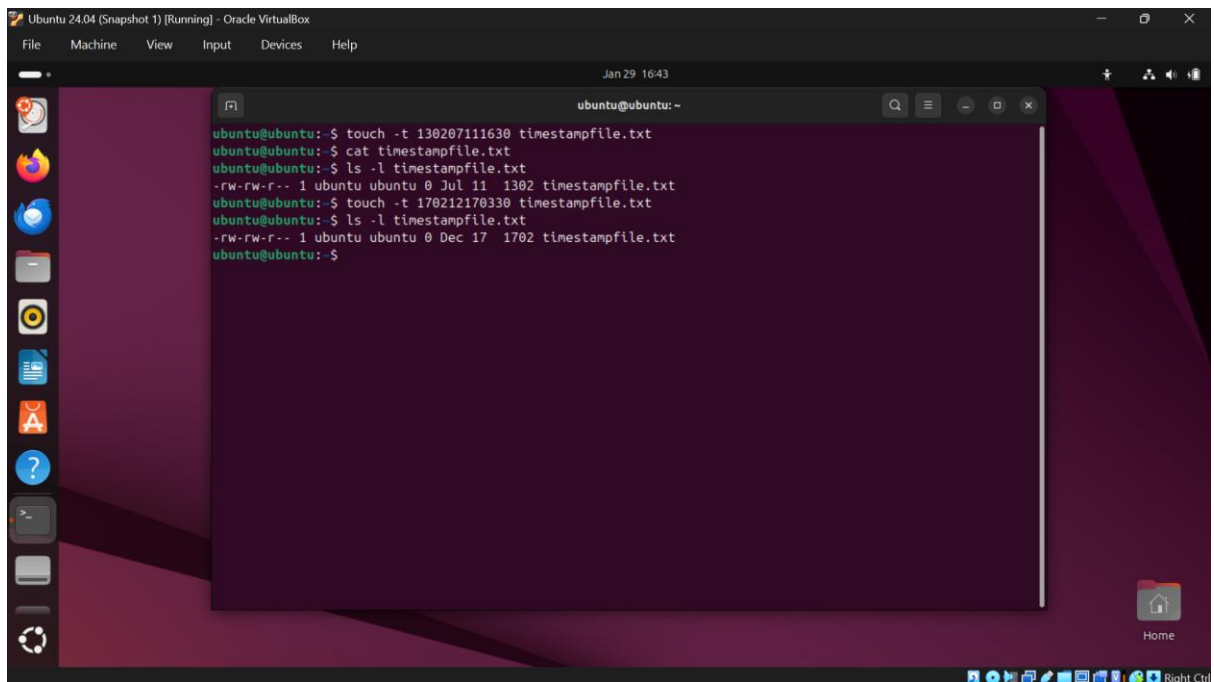
Ubuntu 24.04 (Snapshot 1) [Running] - Oracle VirtualBox

File Machine View Input Devices Help

Jan 29 18:47

ubuntu@ubuntu: ~

```
ubuntu@ubuntu:~$ touch file1.txt
ubuntu@ubuntu:~$ mv file1.txt renamedfile.txt
ubuntu@ubuntu:~$ ls
24K-8579  Documents  Music  Pictures  Templates  ab.txt  dummy.txt  hecopy.me  mecopy  pipercopy  renamedfile.txt
Desktop  Downloads  OS     Public   Videos   ab212.txt  hecopy    letter.txt  mecopy.me  pipercopy.me  snap
ubuntu@ubuntu:~$
```



Ubuntu 24.04 (Snapshot 1) [Running] - Oracle VirtualBox

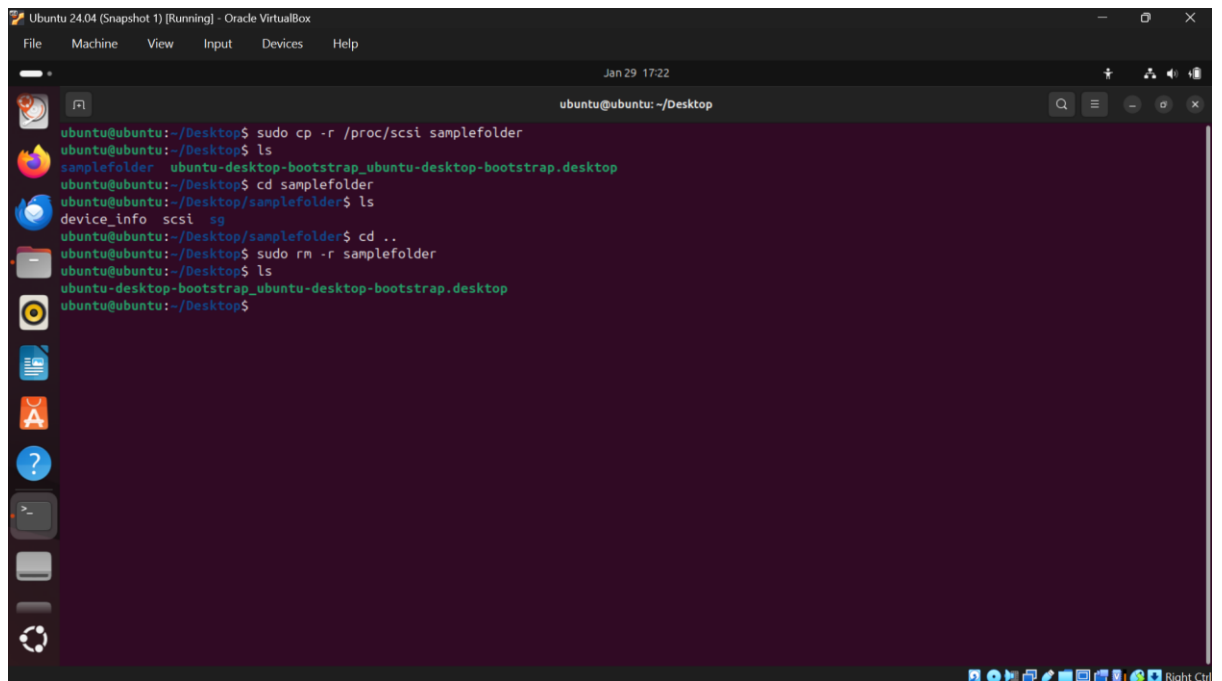
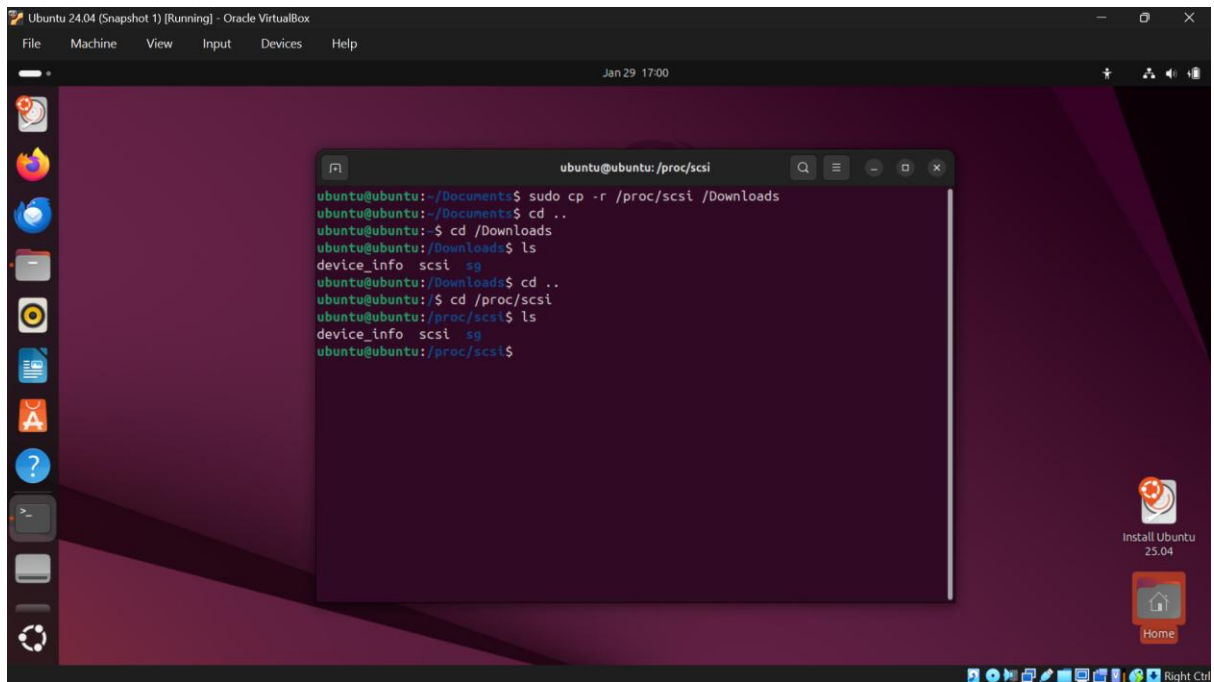
File Machine View Input Devices Help

Jan 29 16:43

ubuntu@ubuntu: ~

```
ubuntu@ubuntu:~$ touch -t 130207111630 timestampfile.txt
ubuntu@ubuntu:~$ cat timestampfile.txt
ubuntu@ubuntu:~$ ls -l timestampfile.txt
-rw-rw-r-- 1 ubuntu ubuntu 0 Jul 11 1302 timestampfile.txt
ubuntu@ubuntu:~$ touch -t 170212170330 timestampfile.txt
ubuntu@ubuntu:~$ ls -l timestampfile.txt
-rw-rw-r-- 1 ubuntu ubuntu 0 Dec 17 1702 timestampfile.txt
ubuntu@ubuntu:~$
```

h) /proc directory contains too many folders and subfolder making it time and resource intensive to copy and crashes terminal an VM, so I copied a file and smaller folder instead



```
Ubuntu 24.04 (Snapshot 1) [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Jan 29 17:44
ubuntu@ubuntu: -
ubuntu@ubuntu:~$ sudo adduser abc
info: Adding user 'abc' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group 'abc' (1002) ...
info: Adding new user 'abc' (1002) with group 'abc (1002)' ...
info: Creating home directory '/home/abc' ...
info: Copying files from '/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for abc
Enter the new value, or press ENTER for the default
  Full Name []: abc
    Room Number []:
    Work Phone []:
    Home Phone []:
      Other []:
Is the information correct? [Y/n] y
info: Adding new user 'abc' to supplemental / extra groups 'users' ...
info: Adding user 'abc' to group 'users' ...
ubuntu@ubuntu:~$ touch dummy.txt
ubuntu@ubuntu:~$ ls -l dummy.txt
-rw-rw-r-- 1 ubuntu ubuntu 0 Jan 29 17:41 dummy.txt
ubuntu@ubuntu:~$ sudo chown abc dummy.txt
ubuntu@ubuntu:~$ ls -l dummy.txt
-rw-rw-r-- 1 abc ubuntu 0 Jan 29 17:41 dummy.txt
ubuntu@ubuntu:~$
```

i)Killing the process 4049 will close terminal

```
Ubuntu 24.04 (Snapshot 1) [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Jan 29 17:49
ubuntu@ubuntu: -
ubuntu@ubuntu:~$ ps -A
  PID TTY          TIME CMD
    1 ?        00:00:02 systemd
    2 ?        00:00:00 kthreadd
    3 ?        00:00:00 pool_workqueue_release
    4 ?        00:00:00 kworker/R-rcu_gp
    5 ?        00:00:00 kworker/R-sync_wq
    6 ?        00:00:00 kworker/R-kvfree_rcu_reclaim
    7 ?        00:00:00 kworker/R-slub_flushwq
    8 ?        00:00:00 kworker/R-netns
    9 ?        00:00:01 kworker/0:0-events
   11 ?        00:00:00 kworker/0:0H-events_highpri
   13 ?        00:00:00 kworker/R-mm_percpu_wq
   14 ?        00:00:00 rcu_tasks_kthread
   15 ?        00:00:00 rcu_tasks_rude_kthread
   16 ?        00:00:00 rcu_tasks_trace_kthread
   17 ?        00:00:01 ksoftirqd/0
   18 ?        00:00:03 rcu_preempt
   19 ?        00:00:00 rcu_exp_par_gp_kthread_worker/0
   20 ?        00:00:00 rcu_exp_gp_kthread_worker
   21 ?        00:00:00 migration/0
   22 ?        00:00:00 idle_inject/0
   23 ?        00:00:00 cpuhp/0
   24 ?        00:00:00 kdevtmpfs
   25 ?        00:00:00 kworker/R-inet_frag_wq
   26 ?        00:00:00 kauditd
   27 ?        00:00:00 khungtaskd
```

```
Ubuntu 24.04 (Snapshot 1) [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Jan 29 17:49
ubuntu@ubuntu: ~
3441 ? 00:00:00 subiquity-serve
3460 ? 00:00:04 python3.12
3484 ? 00:00:00 power-profiles-
3655 ? 00:00:00 kworker/R-xfsalloc
3656 ? 00:00:00 kworker/R-xfs_mru_cache
3659 ? 00:00:00 jfsIO
3660 ? 00:00:00 jfsCommit
3661 ? 00:00:00 jfsSync
3833 ? 00:00:00 kworker/0:1-ata_sff
3925 ? 00:00:00 kworker/u4:0-loop0
3938 ? 00:00:02 nautilus
3962 ? 00:00:00 systemd-hostnam
3963 ? 00:00:00 gvfsd-recent
3964 ? 00:00:00 gvfsd-network
3976 ? 00:00:00 gvfsd-dnssd
3982 ? 00:00:00 gvfsd-wsdd
3987 ? 00:00:00 python3
3989 ? 00:00:00 uuidd
4000 ? 00:00:00 gnome-control-c
4002 ? 00:00:00 gnome-calculato
4005 ? 00:00:00 gnome-character
4006 ? 00:00:00 gnome-clocks
4049 ? 00:00:00 gnome-terminal-
4056 pts/0 00:00:00 bash
4063 pts/0 00:00:00 ps
ubuntu@ubuntu:~$ kill 4049
```

```
Ubuntu 24.04 (Snapshot 1) [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Jan 29 17:53
ubuntu@ubuntu: ~
ubuntu@ubuntu:~$ sudo find / -name "*lab"
/proc/spl/kmem/slab
find: '/run/user/1000/doc': Permission denied
find: '/run/user/1000/gvfs': Permission denied
/sys/kernel/slab
/sys/kernel/slab/kmalloc-rnd-01-16/objs_per_slab
/sys/kernel/slab/memdup_user-4k/objs_per_slab
/sys/kernel/slab/kmalloc-rnd-02-512/objs_per_slab
/sys/kernel/slab/kmalloc-rnd-13-128/objs_per_slab
/sys/kernel/slab/kmalloc-rnd-12-4k/objs_per_slab
/sys/kernel/slab/zfs_znode_hold_cache/objs_per_slab
/sys/kernel/slab/:0001328/objs_per_slab
/sys/kernel/slab/TCPv6/objs_per_slab
/sys/kernel/slab/:0000560/objs_per_slab
/sys/kernel/slab/zap_attr_long_cache/objs_per_slab
/sys/kernel/slab/to_kiocb/objs_per_slab
/sys/kernel/slab/kmalloc-rnd-14-96/objs_per_slab
/sys/kernel/slab/:A-0000184/objs_per_slab
/sys/kernel/slab/:0000136/objs_per_slab
/sys/kernel/slab/kmalloc-rnd-06-192/objs_per_slab
/sys/kernel/slab/kmalloc-rnd-15-32/objs_per_slab
/sys/kernel/slab/kmalloc-rnd-06-512/objs_per_slab
/sys/kernel/slab/kmalloc-rnd-11-96/objs_per_slab
/sys/kernel/slab/kmalloc-rnd-15-2k/objs_per_slab
/sys/kernel/slab/dmu_buf_impl_t/objs_per_slab
/sys/kernel/slab/kmalloc-rnd-11-8/objs_per_slab
/sys/kernel/slab/:a-0000168/objs_per_slab
/sys/kernel/slab/kmalloc-rnd-07-4k/objs_per_slab
/sys/kernel/slab/RAWv6/objs_per_slab
/sys/kernel/slab/zap_name_long/objs_per_slab
/sys/kernel/slab/:0000192/objs_per_slab
```

Ubuntu 24.04 (Snapshot 1) [Running] - Oracle VirtualBox

File Machine View Input Devices Help

Jan 29 17:56

ubuntu@ubuntu: ~

```
ubuntu@ubuntu:~$ df
Filesystem            1K-blocks    Used Available Use% Mounted on
tmpfs                  201472      1956    199516   1% /run
/dev/sr0              6131368 6131368     0 100% /cdrom
/cow                  1007352    61532   945820   7% /
tmpfs                  1007352      8    1007344   1% /dev/shm
tmpfs                   5120       8     5112   1% /run/lock
tmpfs                  1024       0     1024   0% /run/credentials/systemd-journald.service
tmpfs                  1024       0     1024   0% /run/credentials/systemd-resolved.service
tmpfs                 1007352      8    1007344   1% /tmp
tmpfs                 201468     160    201308   1% /run/user/1000

ubuntu@ubuntu:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs           197M  2.0M  195M   1% /run
/dev/sr0        5.9G  5.9G   0 100% /cdrom
/cow            984M   61M  924M   7% /
tmpfs           984M  8.0K  984M   1% /dev/shm
tmpfs           5.0M  8.0K   5.0M   1% /run/lock
tmpfs           1.0M   0 1.0M   0% /run/credentials/systemd-journald.service
tmpfs           1.0M   0 1.0M   0% /run/credentials/systemd-resolved.service
tmpfs           984M  8.0K  984M   1% /tmp
tmpfs           197M 160K  197M   1% /run/user/1000
```

TASK 5:

Ubuntu 24.04 (Snapshot 1) [Running] - Oracle VirtualBox

File Machine View Input Devices Help

Jan 29 18:13

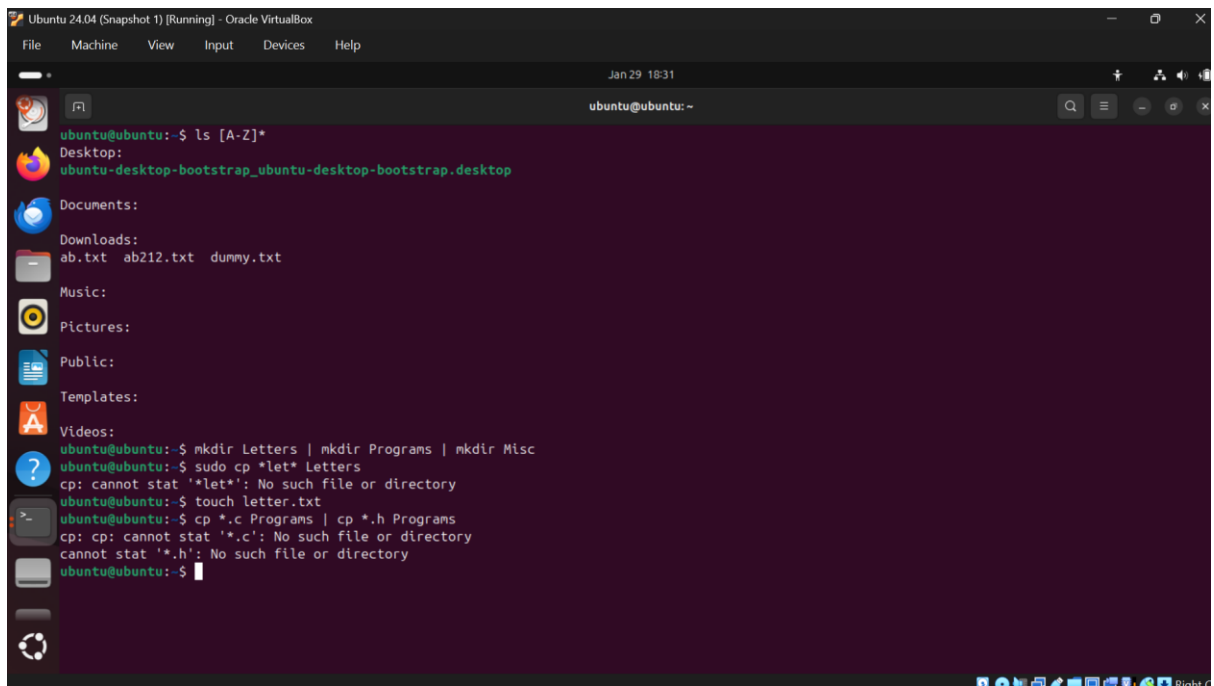
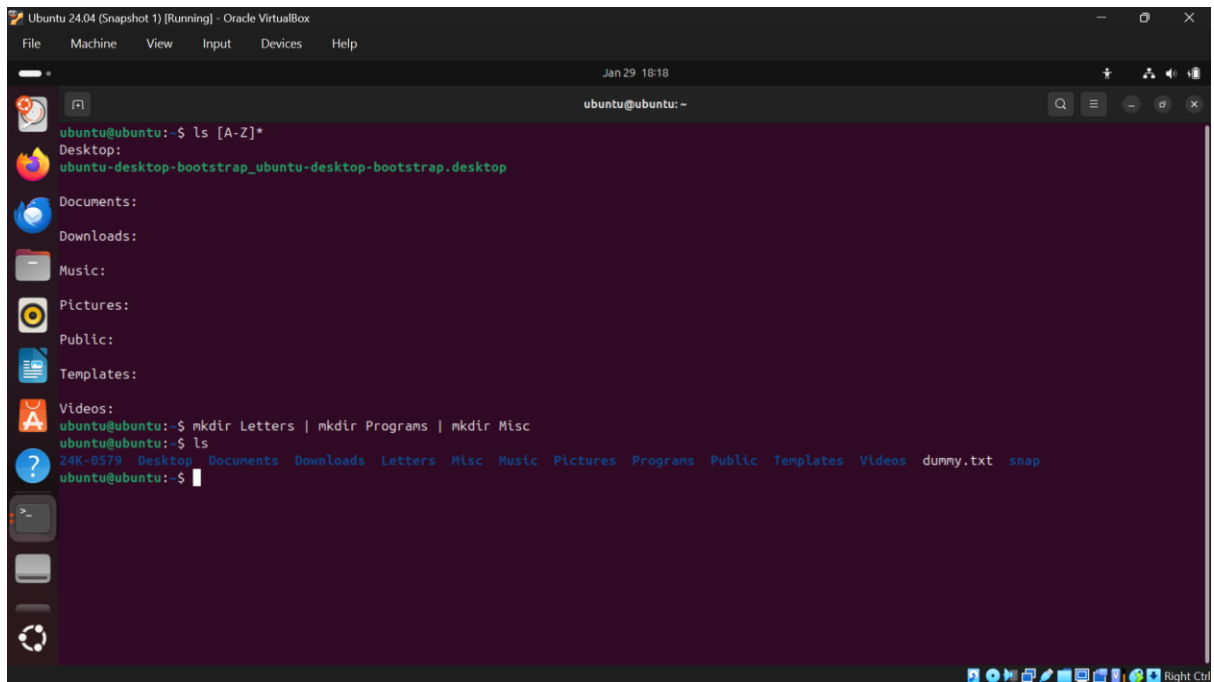
ubuntu@ubuntu: /home

```
ubuntu@ubuntu:~$ ls bin/*sh
bin/bash      bin/chsh      bin/dash      bin/instmodsh  bin/sh      bin/static-sh  bin/xrefresh
bin/c_rehash  bin/crash     bin/gettext.sh  bin/rbash      bin/ssh      bin/tclsh
```

ubuntu@ubuntu:~\$ cd /home/24K-0579

ubuntu@ubuntu:/home/24K-0579\$ cd ..

ubuntu@ubuntu:/home\$ cd




```
Ubuntu 24.04 (Snapshot 1) [Running] - Oracle VirtualBox
File Machine View Input Devices Help

Jan 29 18:37
ubuntu@ubuntu: ~/OS

ubuntu@ubuntu:~$ mkdir OS
ubuntu@ubuntu:~$ touch pipercopy.me mecapy.me hecopy.me
ubuntu@ubuntu:~$ cp *copy.me OS
ubuntu@ubuntu:~$ touch pipemove.me menove.me henove.me
ubuntu@ubuntu:~$ mv *move.me OS
ubuntu@ubuntu:~$ ls
24K-0579 Documents Letters Music Pictures Public Videos ab212.txt hecopy letter.txt mecapy mecopy pipercopy.me
Desktop Downloads Misc OS Programs Templates ab.txt dummy.txt hecopy.me mecapy pipercopy snap
ubuntu@ubuntu:~$ cd OS
ubuntu@ubuntu:~/OS$ ls
hecopy.me henove.me mecapy.me menove.me pipemove.me pipercopy.me
ubuntu@ubuntu:~/OS$ touch abdelcs delase accdel
ubuntu@ubuntu:~/OS$ ls
abdelcs accdel delase hecopy.me henove.me mecapy.me menove.me pipemove.me pipercopy.me
ubuntu@ubuntu:~/OS$ rm *del*
ubuntu@ubuntu:~/OS$ ls
hecopy.me henove.me mecapy.me menove.me pipemove.me pipercopy.me
ubuntu@ubuntu:~/OS$
```